

Dynamics Lab

Abstract

The Young's modulus of a stainless-steel ruler was determined using vibration. The ruler was set up as a cantilever, and its vibrations were recorded using a microphone module and a Pico microcontroller. The frequency of vibration was collected from the recorded data by Pico using zero-crossing analysis, and Young's modulus was calculated for different lengths. The average values were approximately 201, 207 and 216 GPa for each length which is close to expected value for stainless steel, around 200GPa. Overall, the result shows that this method provides a reasonable estimation for material Young modulus.

1. Introduction

The stiffness of a material can be described by its Young's modulus. This is an important property in engineering as it shows how material components behave under force (WIKIPEDIA, 2026).

In this experiment, the Young's modulus of a stainless-steel ruler is estimated by measuring its vibration when set up as a cantilever. The oscillation frequency is obtained from recorded signals and used to calculate the material property for different overhang lengths. When a cantilever beam is displaced and released it vibrates at a natural frequency that is different for all materials. From this relationship we can calculate expected Young modulus using its frequency.

This type of measurements can be useful in situations where material properties need to estimated quickly without advanced testing equipments (like in emergency situations) (University of Bristol, 2026).

2. Methods

This section describes the equipment used and the procedure followed to investigate the Young's modulus of a stainless-steel ruler by acquiring signals with a microphone circuit, plotting the data, and performing the necessary calculations.

2.1. Apparatus

The experimental setup consisted of a stainless-steel ruler clamped to the edge of a table, a microphone sensor connected to a custom pre-amplifier circuit, and a Pico microcontroller for data acquisition. The microphone circuit was built on a breadboard using a condenser microphone element, variable resistors and capacitors, a trimmer potentiometer, Zener and Schottky diodes, 9V power supplies, and an LM741 operational amplifier. The microphone circuit used in the experiment is shown in Figure 1. A laptop running Python code for the Pico was used to record the vibration signal at 5kHz. The stainless-steel ruler used in the experiment had a total length of 0.30 m, width of 0.025 m, and thickness of 0.001 m. The mass of the ruler was approximately 0.060 kg (Dortech, 2026). These values were used to calculate the second moment of area and Young's modulus. The overall experimental setup is shown Figure 2.

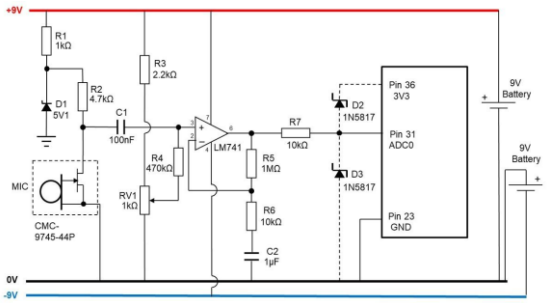


Figure 1 - Microphone circuit diagram

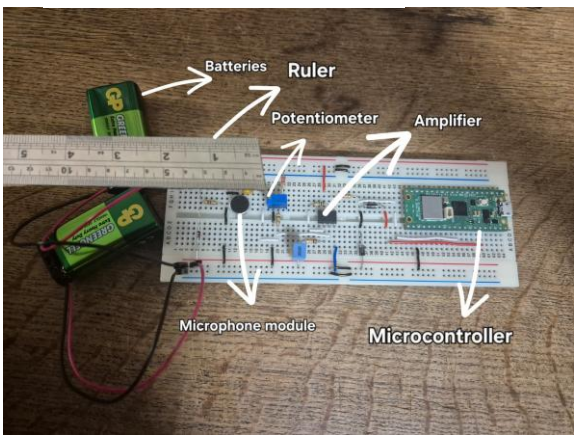


Figure 2 - Experimental setup

2.2. Preparation

The stainless-steel ruler was positioned on the table with one end extended by a determined length, and a heavy object such as a book was placed on top of the ruler to provide stable support. The condenser microphone circuit assembled on a breadboard was placed 5–6 cm below the free end of the ruler to ensure a clear signal without and background noise, and the microcontroller was connected to the laptop where the provided signal-acquisition Python code was opened and prepared for recording.

2.3. Experiment Procedure

For each measurement, the signal-acquisition Python code was executed on the laptop to begin recording. The stainless-steel ruler was set to an overhang of 10 cm, and the free end was vibrated and left to oscillate while the signal was acquired. This process was repeated three times for the 10 cm length. The same steps were then carried out for overhang lengths of 12 cm and 14 cm to reduce errors of random variation. All recordings were saved for later analysis of the vibration frequency and Young's modulus.

2.4. Data Analysis

The recorded signals were saved as CSV files and imported into Python for processing. Each signal was plotted, and the DC component was removed so that the waveform was centred around zero. The vibration region of the signal was then cropped to remove background noise, and the zero-crossing points were identified to allow the oscillation frequency to be calculated. For each overhang length, the three measured frequencies were averaged, and these values were used to calculate the Young's modulus using the cantilever vibration equation (1.

3. Results

The vibration signals obtained from the experiment were analysed to determine the oscillation frequency and calculate the Young's modulus for each overhang length. Figure 3 shows an example of the raw signal before processing, illustrating the presence of noise and the need for cropping.

The cropped results for all trials are summarised in the tables and shown in Figures 4-6, Figure 4 shows the signal for 10 cm, Figure 5 for 12 cm, and Figure 6 for 14 cm (for clarity, only one representative cropped signal is presented for each overhang length, although three trials were conducted per each length.)

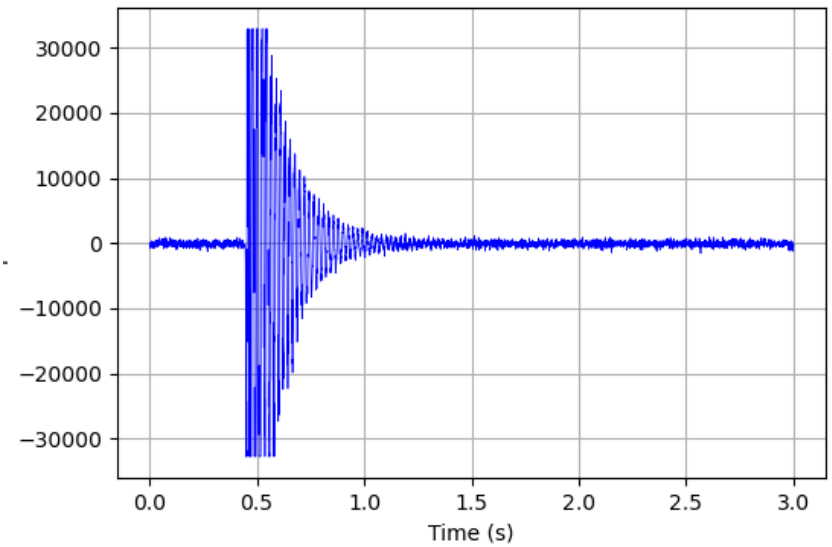


Figure 3 – Raw signal of 10cm overhang (TRIAL 1)

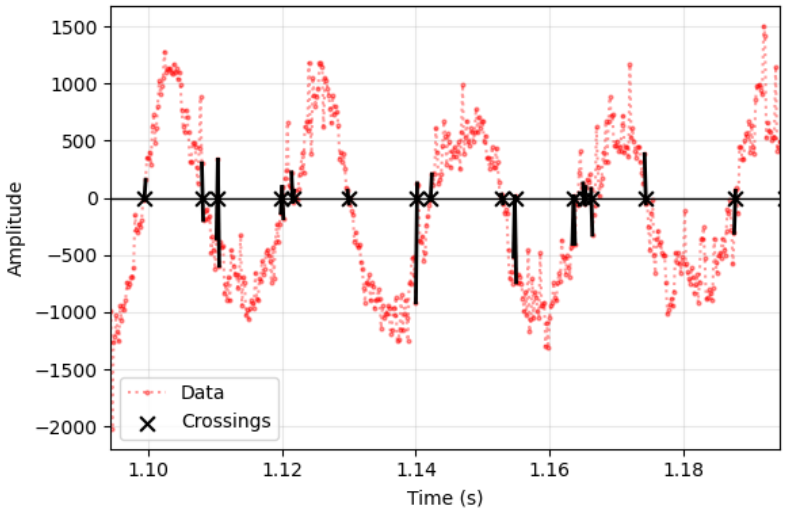


Figure 4 - Cropped data of 10cm overhang (TRIAL 1)

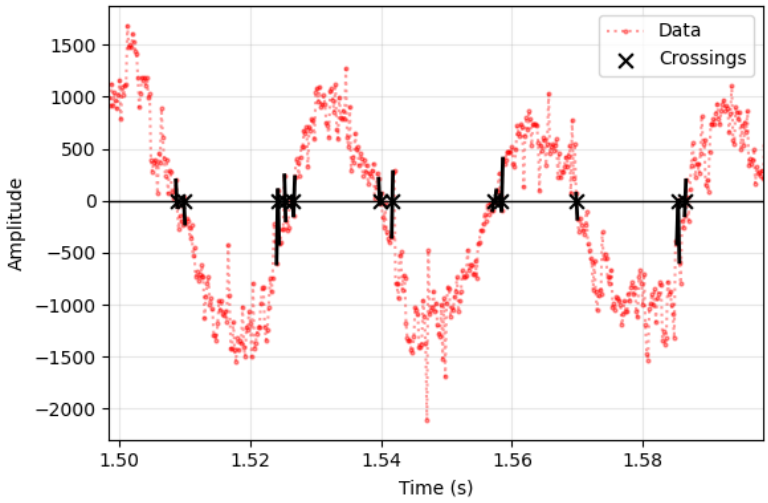


Figure 5 - Cropped data of 12cm overhang (TRIAL 2)

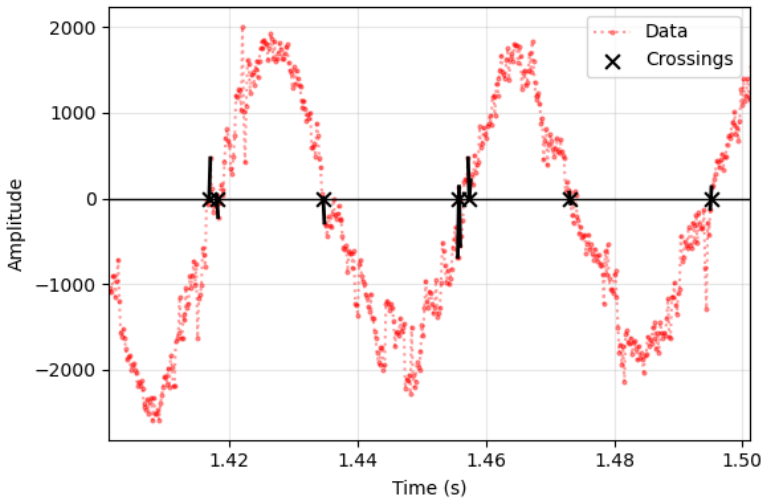


Figure 6 - Cropped data of 14cm overhang (TRIAL 3)

Table 1 shows the measured frequencies(in 2 sf) for each overhang length and trial.

Table 1 - Measured frequencies for each overhang length and trial.

Length	Repeat	Frequency (Hz)
10cm	1	88
	2	79
	3	89
12cm	1	59
	2	67
	3	53
14cm	1	45
	2	36
	3	52

The Young's modulus was calculated using the cantilever vibration *equation* (1 (University of Bristol, 2026):

$$f = 0.56\sqrt{\frac{EI}{qL^4}} \tag{1}$$

where f is frequency (Hz), E is Young's modulus (Pa), I is second moment of area (m⁴), q is mass per unit length ($\frac{kg}{m}$), and L is overhang length (m).

$$E = \frac{(f/0.56)^2 qL^4}{I} \tag{2}$$

where $I = \frac{bh^3}{12}$ is the 2nd moment of area for a rectangular cross section (Engineering Tool Box, 2026), where *b* is width (m) and *h* is thickness (m).,

The values were used to determine E in Gpa. The average Young's modulus for each length was calculated and shown in Table 2.

Table 2 - Calculated Young's modulus values and averages for each overhang length

Length	Repeat	Young's Modulus E (GPa)	Average Young's Modulus E (GPa)
10cm	1	211	201
	2	173	
	3	217	
12cm	1	198	207
	2	261	
	3	160	
14cm	1	217	216
	2	143	
	3	288	

The oscillation frequency decreases with increasing overhang length, while the calculated Young's modulus values remain broadly consistent across all lengths.

4. Discussion

The results show that frequency decreases as the overhang length increases, which is consistent with the expected behavior of a cantilever beam. This is also in agreement with the theoretical relationship $f \propto \frac{1}{L^2}$. The calculated Young's modulus values are similar across all lengths and are close to the expected value for stainless steel, which is approximately 200 GPa (AZoM, 2026)

There is some variation between the individual trials, with certain values differing from the average. This may be caused by experimental uncertainties such as inconsistent manual vibration of the ruler, variation in the force used to release the ruler between trials and different overhang lengths, and background noise affecting the recorded signal. These factors can influence the amplitude and quality of the signal, leading to variation in the measured frequency. In addition, the surface used to fix the ruler may not have been completely rigid or stable, which could affect the boundary condition of the cantilever and lead to small changes in the measured frequency.

It can also be observed that the results for the 14 cm overhang show greater variation compared to the shorter lengths. This is likely because the longer overhang produces lower frequency oscillations, which are more affected by background noise and measurement uncertainty. As a result, the calculated Young's modulus values for 14 cm show a wider spread.

In addition, inaccuracies in measuring the ruler dimensions, particularly the thickness, can significantly affect the calculated value of Young's modulus, as thickness is cubed in the second moment of area calculation. Although the results are close to the expected value, there are still sources of uncertainty that could be improved. The thickness of the ruler could be measured more precisely using calipers to reduce measurement error. A fixed method of releasing the ruler could also be used instead of manual vibration to ensure more consistent oscillations. The identified errors mainly introduce variation in the measured frequency, which leads to variation in the calculated Young's modulus values, and may cause the results to be either overestimated or underestimated.

5. Conclusion

The Young's modulus of a stainless-steel ruler was successfully determined using vibration analysis. The oscillation frequency was measured for different overhang lengths, and the calculated values of Young's modulus were found to be consistent and close to the expected value of approximately 200 GPa. Although some variation was observed between trials, the results were of the correct order of magnitude, indicating that the method provides a reasonable estimate of the material's stiffness.

Future work could improve the accuracy of the results by using more precise measurement tools for the ruler dimensions and a controlled method for releasing the ruler to ensure more consistent oscillations.

6. References

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